Problem Set: Area Between Curves

Area between curves, polar areas, and the Gini index

Name: _____ **Date:** _____

Directions: Show all work on separate paper. For each problem, sketch the region (unless a graph is given), find any points of intersection, set up the integral, and evaluate it exactly.

Warm-Up

1. Find the area of the region bounded by $y = x^2$ and $y = 2x + 3$. Integrate with respect to x.
2. Find the area of the region bounded by $x = y^2$ and $y = x - 2$. Integrate with respect to y.

Part I: Area Between Curves

3. Find the total area enclosed by $y = \dfrac{x}{x^2 + 1}$ and $y = \dfrac{x}{2}$. (The region consists of two pieces.)
4. Find the area between $y = \cos x$ and $y = \cos^3 x$ on $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$.
5. Find the area between $y = \sin^2 x$ and $y = \sin^4 x$ on $[0, \pi]$.
6. Find the area of the region bounded by $y = \dfrac{1}{x+1}$, $y = \dfrac{1}{x(x+1)}$, and $x = 3$.
7. Find the area enclosed by $y = x$ and $y = x \ln x$ for $x > 0$.

Part II: The Gini Index. A Lorenz curve $y = L(x)$ gives the fraction $L(x)$ of a population's total income earned by the poorest fraction x of the population. The Gini index is twice the area between the line of equality $y = x$ and the Lorenz curve:

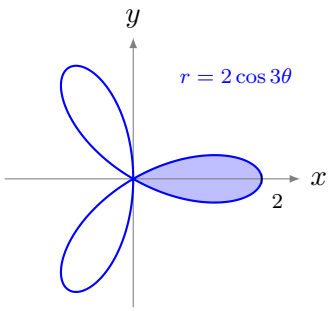
$$G = 2 \int_0^1 \big(x - L(x)\big)\, dx.$$

$G = 0$ means perfect equality, and values closer to 1 mean greater inequality. Sketch each Lorenz curve together with $y = x$ and shade the region.

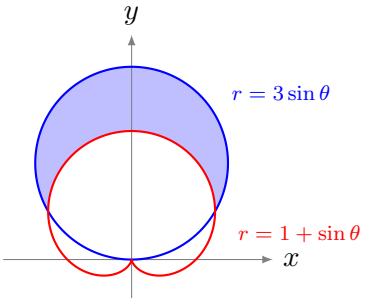
8. A country's Lorenz curve is $L(x) = x\, e^{x-1}$ for $0 \le x \le 1$. (a) Verify that $L(0) = 0$ and $L(1) = 1$. (b) Find the Gini index exactly, then give a decimal approximation.
9. A second country's Lorenz curve is $L(x) = 1 - \sqrt{1 - x^2}$ for $0 \le x \le 1$. (a) Find the Gini index exactly, then give a decimal approximation. (b) Which country from Problems 8 and 9 has the more unequal income distribution?

Part III: Polar Areas. Each region is shaded on its graph. Find the points of intersection (where needed), set up the integral, and evaluate it exactly.

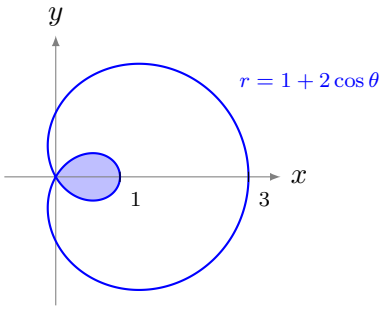
10. Find the area of one petal of the rose $r = 2\cos(3\theta)$.



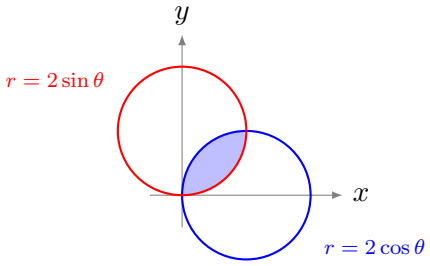
11. Find the area inside $r = 3\sin\theta$ and outside $r = 1 + \sin\theta$.



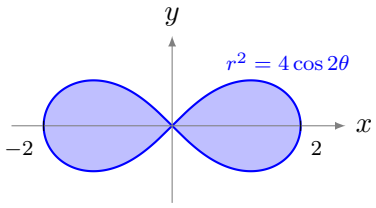
12. Find the area of the inner loop of the limaçon $r = 1 + 2\cos\theta$.



13. Find the area inside both $r = 2\cos\theta$ and $r = 2\sin\theta$.



14. Find the total area enclosed by the lemniscate $r^2 = 4\cos(2\theta)$.

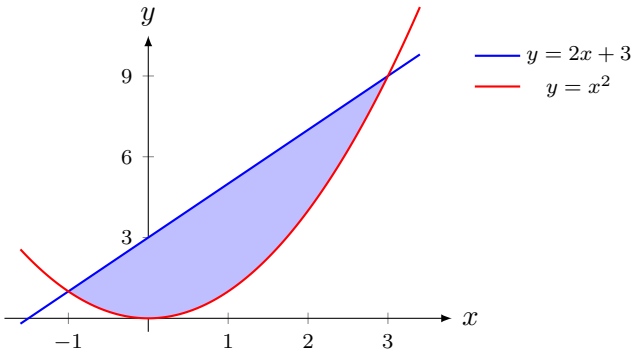


Solutions

Warm-Up

1. **In x.** $x^2 = 2x + 3 \Rightarrow (x-3)(x+1) = 0$, so $x = -1, 3$. The line is on top.

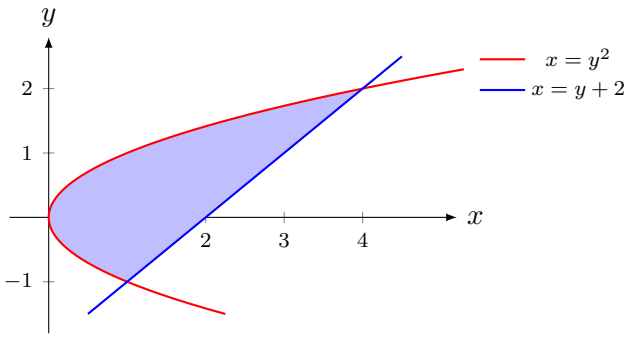
$$A = \int_{-1}^{3} \left(2x + 3 - x^2\right) dx = \left[x^2 + 3x - \tfrac{x^3}{3}\right]_{-1}^{3} = 9 - \left(-\tfrac{5}{3}\right) = \frac{32}{3}.$$



2. **In y.** Rewrite the line as $x = y + 2$. Then $y^2 = y + 2 \Rightarrow (y-2)(y+1) = 0$, so $y = -1, 2$. The line is on the right.

$$A = \int_{-1}^{2} \left(y + 2 - y^2\right) dy = \left[\tfrac{y^2}{2} + 2y - \tfrac{y^3}{3}\right]_{-1}^{2} = \tfrac{10}{3} - \left(-\tfrac{7}{6}\right) = \frac{9}{2}.$$

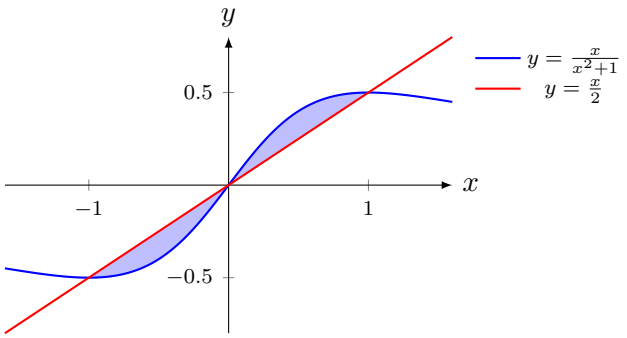
(In x, this region would need two integrals, split at $x = 1$.)



Part I: Area Between Curves

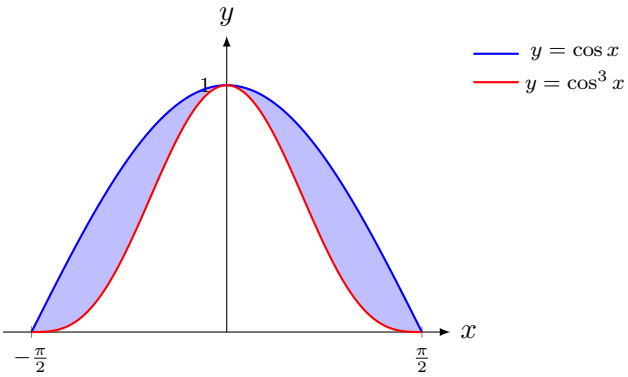
3. **Substitution.** The curves meet at $x = -1, 0, 1$. By odd symmetry,

$$A = 2\int_0^1 \left(\frac{x}{x^2+1} - \frac{x}{2}\right) dx = \ln 2 - \tfrac{1}{2}.$$



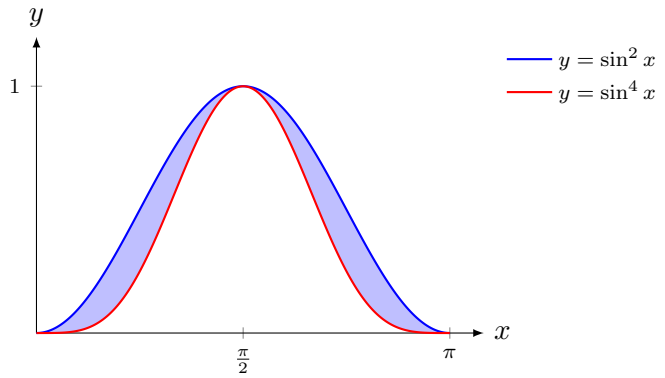
4. **Trig powers.** $\cos x - \cos^3 x = \cos x \sin^2 x$; let $u = \sin x$.

$$A = \int_{-\pi/2}^{\pi/2} \cos x \sin^2 x \, dx = \tfrac{2}{3}.$$



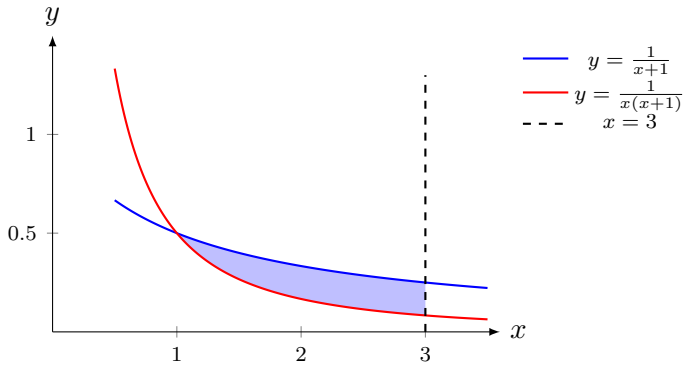
5. **Trig powers.** $\sin^2 x - \sin^4 x = \sin^2 x \cos^2 x = \frac{1}{4}\sin^2(2x)$.

$$A = \int_0^{\pi} \tfrac{1}{4}\sin^2(2x)\,dx = \frac{\pi}{8}.$$



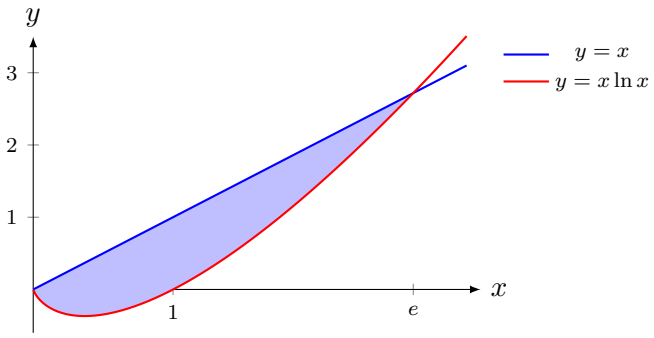
6. **Partial fractions.** The curves meet at $x = 1$, and $\dfrac{1}{x+1} - \dfrac{1}{x(x+1)} = \dfrac{x-1}{x(x+1)} = -\dfrac{1}{x} + \dfrac{2}{x+1}$.

$$A = \int_1^3 \left(-\frac{1}{x} + \frac{2}{x+1}\right) dx = \ln\frac{4}{3}.$$



7. **Integration by parts.** The curves meet at $x = e$; use $\lim_{x\to 0^+} x^2 \ln x = 0$.

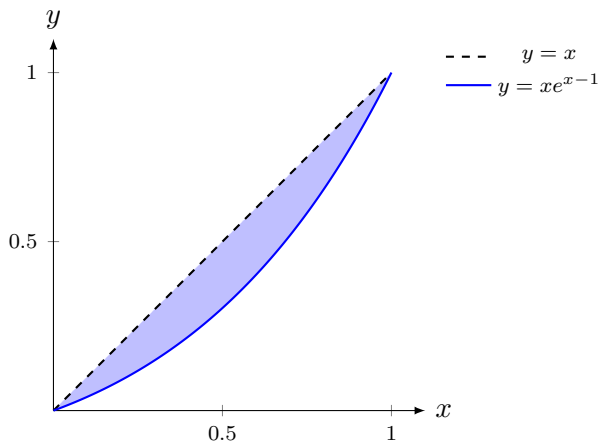
$$A = \int_0^e (x - x\ln x)\,dx = \frac{e^2}{4}.$$



Part II: The Gini Index

8. **Integration by parts.** (a) $L(0) = 0 \cdot e^{-1} = 0$ and $L(1) = 1 \cdot e^{0} = 1$. (b) With $u = x$, $dv = e^{x-1}dx$: $\int_0^1 xe^{x-1}\,dx = \left[(x-1)e^{x-1}\right]_0^1 = \frac{1}{e}$.

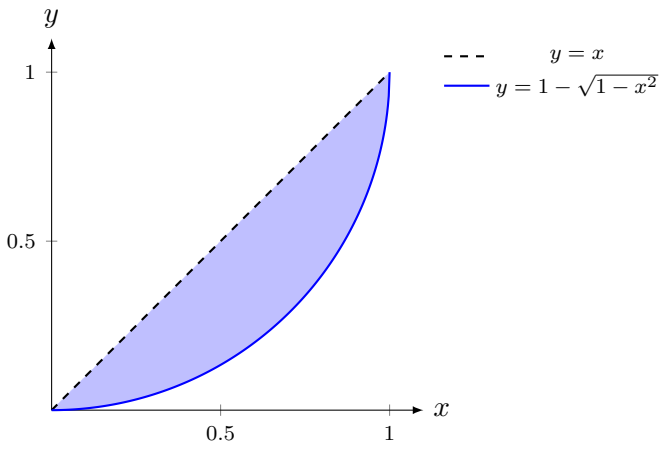
$$G = 2\int_0^1 \left(x - xe^{x-1}\right) dx = 2\left(\frac{1}{2} - \frac{1}{e}\right) = 1 - \frac{2}{e} \approx 0.264.$$



9. **Trig substitution.** (a) With $x = \sin\theta$, $\int_0^1 \sqrt{1-x^2}\,dx = \int_0^{\pi/2} \cos^2\theta\,d\theta = \frac{\pi}{4}$, so $\int_0^1 L(x)\,dx = 1 - \frac{\pi}{4}$.

$$G = 2\left(\frac{1}{2} - \left(1 - \frac{\pi}{4}\right)\right) = \frac{\pi}{2} - 1 \approx 0.571.$$

(b) The second country is more unequal, since $0.571 > 0.264$.



Part III: Polar Areas

10. **Rose petal.** One petal runs between consecutive zeros, $\cos 3\theta = 0$ at $\theta = \pm\frac{\pi}{6}$.

$$A = \frac{1}{2}\int_{-\pi/6}^{\pi/6} 4\cos^2(3\theta)\,d\theta = \int_{-\pi/6}^{\pi/6} \big(1+\cos 6\theta\big)\,d\theta = \frac{\pi}{3}.$$

11. **Between two curves.** $3\sin\theta = 1+\sin\theta \Rightarrow \sin\theta = \frac{1}{2}$, so $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$.

$$A = \frac{1}{2}\int_{\pi/6}^{5\pi/6} \Big[(3\sin\theta)^2 - (1+\sin\theta)^2\Big]d\theta = \frac{1}{2}\int_{\pi/6}^{5\pi/6} \big(8\sin^2\theta - 2\sin\theta - 1\big)\,d\theta = \pi.$$

12. **Inner loop.** $r < 0$ when $\cos\theta < -\frac{1}{2}$, so the inner loop is traced for $\frac{2\pi}{3} \le \theta \le \frac{4\pi}{3}$.

$$A = \frac{1}{2}\int_{2\pi/3}^{4\pi/3} (1+2\cos\theta)^2\,d\theta = \frac{1}{2}\Big[3\theta + 4\sin\theta + \sin 2\theta\Big]_{2\pi/3}^{4\pi/3} = \pi - \frac{3\sqrt{3}}{2}.$$

13. **Overlap of two circles.** The curves meet at the pole and at $\theta = \frac{\pi}{4}$. By symmetry about $\theta = \frac{\pi}{4}$,

$$A = 2\cdot\frac{1}{2}\int_0^{\pi/4} (2\sin\theta)^2\,d\theta = 2\Big[\theta - \tfrac{1}{2}\sin 2\theta\Big]_0^{\pi/4} = \frac{\pi}{2} - 1.$$

14. **Lemniscate.** $r^2 \ge 0$ requires $\cos 2\theta \ge 0$. By symmetry, take four times the area for $0 \le \theta \le \frac{\pi}{4}$:

$$A = 4\cdot\frac{1}{2}\int_0^{\pi/4} 4\cos(2\theta)\,d\theta = 4\Big[\sin 2\theta\Big]_0^{\pi/4} = 4.$$